

Perimeter detection for a 350 mm mobile robot

Time-of-flight, mmWave radar and thermal infrared: a catalogue of every candidate part, the coverage geometry that decides how many of them you need, and what each one can actually tell you about a human, a pet or a box at 1, 3, 5, 8 and 10 feet.

Revision A | research date 2026-09-12 | vendors surveyed: Adafruit, DFRobot, STMicroelectronics, and the wider ToF and lidar market

The answer, first

A 350 mm robot needs **twelve multizone time-of-flight sensors in the perimeter ring, six downward cliff sensors, and eight thermal arrays** to do the whole job the brief describes: avoid every obstacle through 360 degrees, never fall down a stair, know a person is there, and know the difference between a person and a pet. That is nineteen ToF modules and eight people sensors, about USD 909 and about 1.0 A at 5 V.

If the budget only stretches to collision avoidance, the number is **eighteen ToF modules and zero people sensors**, about USD 455. That build is honest about what it is: a bumper made of light. It cannot tell you anything is alive.

	Minimum: collision only	Recommended: collision + animate	Maximum: full classification
ToF modules	18 (12 ring, 6 cliff)	19 (12 ring, 6 cliff, 1 scanner)	24 (16 ring, 6 cliff, 1 head, 1 scanner)
People sensors	0	8 thermal arrays	6 thermal + 4 cameras
Hardware cost	USD 455	USD 909	USD 1,440
Average current at 5 V	724 mA	999 mA	3.8 A
"Will I hit it?"	Yes, 360 deg	Yes, plus dark targets and sunbeams	Yes
"Is it alive?"	No	Yes, to about 2.4 m for a pet, 5 m for a human	Yes, in darkness too
"Human or pet?"	No	By fusion, to about 5 ft	Labelled, to 10 ft

The five things that decide this design

1. Use the horizontal field of view, never the diagonal. Every ST VL53 datasheet leads with a diagonal figure. It is a separately measured number, not the geometric diagonal of the square detection volume, so it cannot be converted by dividing by the square root of two. The VL53L5CX is 63 deg diagonal and **45 deg horizontal**. The VL53L7CX is 90 deg diagonal and **60 deg horizontal**. The VL53L1X is 27 deg diagonal and about **19 deg horizontal**. Sizing a ring from the diagonal understates the count by three to six sensors, and this document's own first-pass research made exactly that error before the verification pass caught it.

2. A ring closes only when $n \times \text{FoV} > 360$, and "exactly 360" is the one value to avoid. At exactly 360 the edge rays of adjacent sensors are parallel, so they never meet at any finite range and the wedge between them is blind forever. Eight VL53L5CX at 45 deg is exactly 360 deg and it does not work.

3. The sensors sit on the rim, not at the centre, so even a closed ring is blind near the robot. On a 175 mm radius the wedges between adjacent sensors close at a range

$$u = r \sin(a) / \sin(a - s) \quad a = \text{half the field of view}, \quad s = \pi / n$$

measured from the platform centre. Twelve VL53L5CX close at 513 mm from the centre, 338 mm past the edge. Sixteen close at 343 mm from the centre, inside the robot's own 350 mm footprint. Both figures appear in this document and they do not disagree: **twelve** is the recommended build, whose near-field wedge is covered by a compliant bumper, and **sixteen** is what it costs to close the wedge inside the footprint and drop the bumper from the safety argument.

These closure figures were derived twice, independently and by different algebra - once by a research agent working from the platform geometry, once by `radar-compare/scripts/geometry.py` solving the bisector intersection directly. They agree to the tenth of a millimetre at every count checked: 513.1 mm for twelve at 45 deg, 343.3 mm for sixteen at 45 deg, 670.4 mm for eight at 60 deg, 1170.9 mm for seven at 60 deg, and 343.0 mm for the thirty-eight single-zone VL53L1X sensors it would take to close a ring at 19.27 deg.

4. Mounting height decides what the robot runs over, and it is not a detail. A ring at 800 mm with a 45 deg vertical field of view cannot see a standing cat at 1 ft or at 3 ft at all - the animal is entirely beneath the beam. A ring at 100 to 200 mm sees everything from a sleeping cat upward, at the price of a floor return in every frame. Put the collision ring low.

5. Classification must never gate the stop. There are 167 to 323 ms between the warning-field edge and the protective-field edge for a target walking toward the robot. Micro-Doppler needs about 250 ms of dwell to reach 75 percent; ST's own people-counting reference uses a 2.1 s window; a thermal frame at 4 Hz is 250 ms by itself. Braking runs on raw geometry at ring frame rate. Classification runs in parallel and changes only what the robot does *after* it has stopped.

What this costs you to ignore

- **24 GHz presence radar is rejected for this robot**, despite being the obvious answer for people sensing. Those modules assume a fixed installation. Their static-clutter rejection is the only thing that makes them useful, and a driving robot gives every static object an apparent velocity, which leaves the filter nothing to reject. They also see through walls, so they report the person in the next room as a perimeter contact. Radar earns its place on this robot only where it can hide behind a printed shell that a ToF sensor cannot see through.
- **Nothing here carries a safety rating.** A certified IEC 61496 Type 3 scanner must prove detection against a 1.8 percent reflectance target inside its protective field. A VL53L8CX does not meet that, which is exactly why a certified scanner sees a black sock and this stack does not. Use the industrial arithmetic; do not claim the rating.
- **No combination in this document tells a cat from a dog without a camera.** Nor does any of it work in direct sunlight above about 5 kLux, which is where ST's published tables stop and direct sun is 100 to 120 kLux. Mirrors defeat all of it.

Two standalone product guides

The operator asked for two parts to be assessed on their own, independent of the ring recommendation. Both chapters read without the rest of the document.

- **ST multizone grid time-of-flight.** The latest grid part as of 2026-09-12 is the **VL53L9CX**, a 54 x 42 zone, 2268-zone stacked imager with MIPI output, announced with mass production from early July 2026. It is a camera module, not a microcontroller peripheral, and it is not a drop-in for anything. The part to actually build with is the **VL53L8CX**.
- **Adafruit 4010** is the **Slamtec RPLIDAR A1M8**, a 360 deg spinning laser scanner at USD 99.95, in stock. It is not a proximity sensor and it is easily confused with the unrelated Vishay VCNL4010. It is the best single "see everything around me" part in its price class and it classifies nothing at all.

How to read the rest

The catalogue chapters list every candidate part with its measurement conditions. The master grid rates each one against an adult, a cat and a 30 mm chair leg at 1, 3, 5, 8 and 10 ft, by a published rule rather than by opinion. The computed appendix carries every geometry table, all of it generated by `radar-compare/scripts/geometry.py` from first principles rather than copied from a vendor page. The last chapter gives a different answer for each of the three robot projects in this repository, because their pin, bus and shell budgets are genuinely different.

Master comparison grid

Every candidate part in one place, rated the same way. The tables in this chapter are generated from `radar-compare/data/sensors.csv` by `radar-compare/scripts/grid.py`. No cell is typed by hand, so no cell can flatter a part the catalogue does not support.

How the capability level is decided

Each cell answers one question: at this distance, against this subject, what can this sensor tell you? The level is computed from the sensor's own published numbers by a fixed rule. The rules are printed here so that a reader who disagrees can argue with the rule rather than with the cell.

Level	What the sensor can tell you
0	nothing
1	presence only
2	obstacle + range
3	living vs not
4	human vs pet
5	full silhouette

The rules, in the order they are applied:

- **Out of range is level 0.** For every optical sensor the rated range is used as published, with its stated target reflectance. Where the catalogue carries a dark-target figure, that one is used instead, because a black sock and a dark-furred cat are the cases that matter.
- **Radar range is scaled by target size, because the physics says so.** The radar range equation makes range proportional to the fourth root of radar cross-section. An adult presents about 0.70 m2 at these frequencies and a cat about 0.012 m2, so a module rated 25 m on a walking adult reaches about 9.0 m on a cat. That scaling is applied to every radar row.
- **A grid imager can only report a shape it can resolve.** The level is set by how many zones or pixels the subject spans across, computed from the centre zone, which is the narrowest. Under half a zone a ToF imager still returns a valid range but nothing else. Under about a third of a pixel a thermal array loses the subject in its own noise.
- **A thermal array and a PIR score 0 against furniture, at every range, forever.** A chair leg is at room temperature. There is no contrast to detect and no amount of processing creates one.
- **A presence radar scores 0 against a motionless inanimate object** unless the catalogue records that it reports static targets. Suppressing motionless returns is what clutter rejection is for; it is the same feature that makes the module immune to a curtain.
- **A 2D scanning lidar sees exactly one horizontal plane.** Anything whose top is below that plane or whose bottom is above it is invisible at every range. The catalogue leaves the scan plane height empty because it is a mounting decision, not a property of the part.
- **Level 4 needs evidence, not optimism.** A vendor radar module reaches level 4 only where its shipped firmware reports an actual class. Telling an adult from a cat on raw range and velocity is a micro-Doppler program someone has to write, and none of these modules exposes the data it would need.

Specifications

Sensor	Vendor / SKU	Technology	USD	FoV H x V	Zones	Rated range	Rate	Interface	Line of sight
VL53L1X	Adafruit 3967	tof-single		27 x 27	1	4.0	50		yes
VL53L5CX	Adafruit 5396	tof-multizone		63 x 63	8x8	4.0	15		yes
RPLIDAR A1M8	Adafruit 4010	tof-lidar-scanning		360 x 0	360	12.0	5.5		yes
C4001 mmWave 25m	DFRobot SEN0610	mmwave-24ghz		100 x 40	1	25.0	10		no
AMG8833 Grid-EYE	Adafruit 3538	thermal-ir-array		60 x 60	8x8	7.0	10		yes

Beam footprint width at each survey distance

Width is $2 \cdot d \cdot \tan(\text{FoV}/2)$ using the **horizontal** field of view. For a scanning lidar the cone width is meaningless, so the figure given is the arc between adjacent samples, which is what actually limits whether a thin object is hit by a beam at all.

Sensor	FoV H	1 ft	3 ft	5 ft	8 ft	10 ft
VL53L1X	27	146	439	732	1171	1464
VL53L5CX	63	374	1121	1868	2989	3736
RPLIDAR A1M8	360	5 (sample pitch)	16 (sample pitch)	27 (sample pitch)	43 (sample pitch)	53 (sample pitch)
C4001 mmWave 25m	100	726	2179	3632	5812	7265

Sensor	FoV H	1 ft	3 ft	5 ft	8 ft	10 ft
AMG8833 Grid-EYE	60	352	1056	1760	2816	3520

Against an adult human, 450 mm wide and 1700 mm tall

Sensor	Technology	1 ft	3 ft	5 ft	8 ft	10 ft
VL53L1X	tof-single	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing	0 nothing
VL53L5CX	tof-multizone	4 human vs pet	3 living vs not	0 nothing	0 nothing	0 nothing
RPLIDAR A1M8	tof-lidar-scanning	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
C4001 mmWave 25m	mmwave-24ghz	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
AMG8833 Grid-EYE	thermal-ir-array	4 human vs pet	3 living vs not	1 presence only	1 presence only	1 presence only

Against a standing cat, 140 mm wide and 250 mm tall

This is the column that separates a specification sheet from a design. Most parts that look capable against an adult collapse here, and they collapse for three different reasons: the cat is too small to fill a zone, too small to return radar energy, or entirely beneath the beam.

Sensor	Technology	1 ft	3 ft	5 ft	8 ft	10 ft
VL53L1X	tof-single	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing	0 nothing
VL53L5CX	tof-multizone	3 living vs not	2 obstacle + range	0 nothing	0 nothing	0 nothing
RPLIDAR A1M8	tof-lidar-scanning	3 living vs not	3 living vs not	2 obstacle + range	2 obstacle + range	2 obstacle + range
C4001 mmWave 25m	mmwave-24ghz	3 living vs not	3 living vs not	3 living vs not	3 living vs not	3 living vs not
AMG8833 Grid-EYE	thermal-ir-array	3 living vs not	1 presence only	0 nothing	0 nothing	0 nothing

Against a 30 mm chair leg

The thin, cold, static obstacle. Thermal and PIR score zero by physics. Presence radar scores zero by design. What is left is the honest list of parts that will stop the robot hitting furniture.

Sensor	Technology	1 ft	3 ft	5 ft	8 ft	10 ft
VL53L1X	tof-single	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing	0 nothing
VL53L5CX	tof-multizone	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing	0 nothing
RPLIDAR A1M8	tof-lidar-scanning	2 obstacle + range	2 obstacle + range	2 obstacle + range	0 nothing	0 nothing
C4001 mmWave 25m	mmwave-24ghz	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range	2 obstacle + range
AMG8833 Grid-EYE	thermal-ir-array	0 nothing	0 nothing	0 nothing	0 nothing	0 nothing

Recommendation: three builds, one per budget

Each tier below is one recommendation, not a menu. Prices are USD as read on 2026-09-12 and exclude tax and shipping. Nothing here has been purchased or physically tested; every figure is from a datasheet or a vendor page, and the ones that are not are marked as estimates.

Tier 1 - collision-ring: do not hit anything

USD 455, 724 mA average at 5 V. This tier answers requirement (a) and nothing else. It has no human channel and no pet channel and it is not honest to claim otherwise.

Qty	Part	Vendor / SKU	Unit	Extended
12	VL53L8CX 8x8 multizone dToF carrier	Pololu 3419	24.95	299.40
6	VL53L4CD 1-1300 mm dToF carrier (cliff)	Pololu 3692	13.95	83.70
2	PCA9548 8-channel I2C mux, STEMMMA QT	Adafruit 5626	6.95	13.90
12	Flexible Qwiic cable, 500 mm	SparkFun PRT-17257	2.75	33.00
6	STEMMA QT cable, 100 mm	Adafruit 4210	0.95	5.70
1	RP2040 concentrator, Pico class	estimate	4.00	4.00
1	Compliant bumper, 20-50 mm closed-cell foam plus 4 microswitches	estimate	15.00	15.00
Total				about 455

Mounting. The twelve ring sensors go at **200 mm** height, boresight **horizontal with no cant**, at **30 deg azimuth spacing**. The six cliff sensors go at **150 mm** with a **23 deg down-cant**, four across the front arc at plus and minus 20 and 60 deg, and two at the rear at 150 and 210 deg.

Why twelve and not eight. The VL53L8CX is 45 deg horizontal. Eight of them total exactly 360 deg, which is the one count that must be avoided: at exactly 360 the edge rays of adjacent sensors are parallel and the wedge between them never closes at any range. Twelve gives 180 deg of designed overlap and closes the wedges 513 mm from the platform centre, 338 mm past the skin. The compliant bumper covers what is left, which is why the bumper is in the bill of materials rather than in a list of nice-to-haves.

Why 200 mm and level. Low enough that a 250 mm standing cat is inside the vertical beam from the skin outward; high enough that the floor does not enter the frame until 483 mm, so the ring is not reporting carpet in every zone.

Tier 2 - collision-ring + animate-layer: the recommendation

USD 909, 999 mA average at 5 V. This is the build to make. It adds requirements (b) and (c), and it closes the two failure modes that would otherwise put the robot on the floor or on the cat: dark targets in a lit room, and glass.

Qty	Part	Vendor / SKU	Unit	Extended
-	Everything in Tier 1			455.00
1	Slamtec RPLIDAR C1 360 deg dToF scanner	DFRobot	69.00	69.00
8	Panasonic AMG8833 8x8 thermal array, STEMMMA QT	Adafruit 3538	44.95	359.60
2	RCWL-1601 ultrasonic ranger	Adafruit 4007	3.95	7.90
1	PCA9548 8-channel I2C mux, third	Adafruit 5626	6.95	6.95
8	STEMMA QT cable, 300 mm	Adafruit 4210	1.25	10.00
Total				about 909

Why the scanner, at 100 mm, and why the C1 rather than the A1

The RPLIDAR C1 is the highest value per dollar in the whole study, because one USD 69 part covers both structural weaknesses of a ToF ring:

- **Dark targets.** The C1 is specified 0.05 to 12 m at 70 percent reflectivity and 0.05 to 6 m

at 10 percent. A black cat on a dark carpet is a 10 percent class target: the ToF ring loses it past about 1.55 m in a lit room and the C1 holds it to 6 m.

- **Ambient light.** The C1 is rated to 40,000 lux. The VL53L8CX is characterised at 0 and 5 kLux

only, and a sunlit patch of floor at roughly 100 kLux reads as *empty space* - a wall of zones reporting "signal rate too low", not noisy numbers. Empty space is the dangerous failure.

- **A lying cat.** A scan plane at 100 mm intersects a cat lying on its side at 120 mm. No fixed ring at 200 mm does.

- **A static background map.** 500 points per revolution is enough to scan-match. Once the robot has a map, anything that does not fit it is dynamic, and dynamic is the strongest animate cue available without a camera. A ToF ring has far too few points to ever do this.

The runner-up, **rejected**, is the RPLIDAR A1 - Adafruit 4010, the part this document also covers as a standalone guide. It is triangulation rather than direct time of flight; its own datasheet qualifies the range as "White objects" with no low-reflectance figure published and says "without direct sunlight exposure"; it is 170 g against 110 g, 5.5 Hz against 10 Hz, and 0.15 m minimum range against 0.05 m. On a 350 mm chassis the A1's blind radius nearly reaches the skin. It costs USD 31 more to be worse on every axis that matters here. Buy the A1 if you already own it or you want the better-documented ROS path; buy the C1 for this robot.

Why eight thermal arrays, at 250 mm, level

Thermal is the **only** modality that separates animate from inanimate, and the only one that sees a *motionless* target. A sleeping cat is invisible to motion-gated radar and invisible to PIR, and plainly visible to thermal.

At 250 mm with a 60 x 60 deg array, the floor first enters frame at 433 mm, the top of a 200 mm cat enters the beam 87 mm from the skin so the animal is in frame essentially always, and a standing adult's head stays in frame to 2.51 m with the torso held beyond that. Ring closure needs $\text{floor}(360/60) + 1 = 7$; eight gives 15 deg of overlap, closes at 670 mm, and leaves a largest hideable sphere of 117.5 mm, smaller than a cat in any dimension.

Rejected: the STHS34PF80 (Adafruit 6426, USD 14.95). A genuinely good part - 10 uA, no Fresnel lens, holds a static human - but it has exactly one sensing element. It reports that something warm is somewhere in an 80 deg cone: no bearing, no size, no contribution to human versus pet. Keep it as a parked-robot wake-up layer only.

Rejected: PIR of every kind, including pet-immune PIR. Pet immunity is a geometric trick - the lens's lowest beam is cut away and the unit is mounted at 2.2 to 2.75 m. At 1.0 m on a robot that elevation separation collapses, and the human's torso and head are now above a downward-fanning lens, so a pet-immune PIR on a low robot can see the pet and miss the person, the exact inversion of its purpose. Worse, PIR responds to the rate of change of flux, so on a driving robot every scene moves and the output is a random number generator.

Two ultrasonic sensors forward, USD 7.90. Glass and mirrors are the one target class the entire optical stack fails on: a ToF sensor ranges to the pane, a lidar returns a dropout or a phantom room, a thermal array sees room-temperature glass. Ultrasonic sees glass reliably. Constrain it hard - warning field only, never in the protective-stop path, and never as the human detector, because soft targets absorb 40 kHz.

The fusion logic

Two rings, both body-fixed, extrinsically calibrated once so a bearing in one is a bearing in the other. The thresholds are a starting point to be calibrated in place.

```
# Tier 0: collision. Geometry alone, at ring frame rate. No classifier in this path.
if min_range_in_swept_corridor < protective_field(v): STOP
elif min_range_in_swept_corridor < warning_field(v): SLOW to 0.3 m/s and announce

# Tier 1.1.4: classification. Runs in parallel. Changes behaviour only AFTER the stop.
match clusters across modalities by bearing (+/-10 deg) and range (+/-150 mm):

if dT < 1.5 K and jitter < 8 mm and planar and world_velocity ~ 0:
OBJECT # box, wall, furniture, door frame

elif dT >= 2.0 K and (h_top > 900 mm or cluster touches the top zone row) and width > 250 mm:
HUMAN # tall AND warm - the high-confidence rule

elif dT >= 1.5 K and h_top < 450 mm and width < 600 mm and (jitter > 8 mm or moving):
PET # low AND warm AND non-rigid. Cat versus dog is NOT attempted.

elif dT >= 1.5 K:
UNKNOWN_LIVING # crouching adult, toddler, large dog -> BEHAVE AS HUMAN

else:
UNKNOWN_OBSTACLE # stop anyway
```

Three rules are embedded in that, and each one is worth more than the code around it:

- **Height is the primary human-versus-pet discriminator**, not temperature and not breathing rate. A resting beagle breathes at 13 to 25 per minute and a resting human at 12 to 20. The distributions overlap almost completely, so a rule of the form `if rate > 20 then pet` will call a calm human a dog. Temperature says *alive*; height says *which*.
- **Never threshold on absolute temperature**. A long-coated dog's flank reads 28.1 degC in a 21 degC room, a 6 K contrast that falls to 2 K in a warm room, while an AMG8833's absolute accuracy is plus or minus 2.5 degC. Apparent temperature is mostly a range measurement in disguise: a cat at 3 m reads 2.5 K and a clothed human at 5 m reads 3.8 K, and moving either by a metre flips the ordering. Use a per-pixel background moving average at about 30 s, which cancels warm-up drift, chassis self-heating and room ambient in one step.
- **The ambiguous class defaults to human**. A crouching adult, a toddler and a Great Dane all land in UNKNOWN_LIVING. Over-classifying as human is the safe failure.

Tier 3 - full-classification-stack

USD 1,440, 3.8 A average at 5 V. Buy this only if labelled classification out to 10 ft is a requirement rather than a wish. The jump from Tier 2 is USD 531 and 2.8 A for one capability.

Qty	Part	Vendor / SKU	Unit	Extended
16	VL53L8CX carrier, 22.5 deg spacing	Pololu 3419	24.95	399.20
6	VL53L4CD carrier (cliff)	Pololu 3692	13.95	83.70
1	VL53L8CH CNH histogram head, 900 mm, own bus	price not published	~35	~35
6	MLX90640-ESF-BAA 32x24 thermal, 110 x 75 deg	Adafruit 4469, out of stock 2026-09-12	74.95	449.70
1	Slamtec RPLIDAR C1	DFRobot	69.00	69.00
2	RCWL-1601 ultrasonic	Adafruit 4007	3.95	7.90
4	PCA9548 8-channel mux	Adafruit 5626	6.95	27.80
1	Raspberry Pi 5	-	80.00	80.00
1	Hailo-8L AI kit	-	70.00	70.00
4	Wide-FoV camera, about 120 deg H, at 900-1000 mm	not priced in any source	~35	~140
-	Concentrator, cabling, bumper	estimate	~79	~79
Total			about 1,440	

Sixteen ToF instead of twelve puts the wedge tip at 343 mm from the centre - 168 mm past the skin, inside the robot's own footprint - and drops the largest sphere that can hide in a seam from 79.4 mm to 56.0 mm. This is the tightest closure worth buying, and it is what lets the compliant bumper stop being part of the safety argument.

What no tier can do, at any range

- **Direct sunlight above about 5 kLux on any optical sensor.** ST publishes 0 and 5 kLux only; the RPLIDAR C1 stops at 40 kLux; direct sun is 100 to 120 kLux. No number exists past that. Do not extrapolate one.
- **Mirrors.** A mirror at 45 deg returns a folded path length and the robot believes the corridor continues. Map known mirrors as permanent keep-out zones.
- **Cat versus dog without a camera.** No combination of ToF, 32x24 thermal and one hobby radar does it.
- **Radar vital signs on a moving robot.** Ego-motion writes a Doppler bias across the scene orders of magnitude larger than a 5 mm chest excursion.
- **Any safety rating.** None of this is IEC 61496 electro-sensitive protective equipment. A certified Type 3 scanner must prove detection at 1.8 percent minimum target reflectance in its protective field; that is the pass mark a VL53L8CX does not meet, and the reason a certified scanner sees a black sock and this stack does not. Use the industrial arithmetic and the Z-factor discipline anyway. Do not claim the rating.

Three decisions that override the parts list, and cost nothing

1. **Make the platform round.** A 350 mm square has a 494.975 mm diagonal and a 72.487 mm corner protrusion. Four 45 deg sensors on the face centres leave the corners permanently uncovered, and moving them to the corners is strictly worse. If it must be square, put the sensors on the face centres, chamfer the corners at 45 deg, and treat the corners as mechanically protected.
2. **Cap speed at 0.5 m/s, and 0.3 m/s in unmapped or occupied space.** A 15 kg robot at 1.0 m/s with a rigid shell delivers a mean 1500 N into a shin against a 130 N ISO/TS 15066 transient limit. The same robot at 0.5 m/s with a 50 mm compliant bumper delivers 38 N. Speed is the cheapest safety component in the build.
3. **Never let classification gate the stop.** Braking runs on raw geometry at frame rate. Classification runs in parallel and only changes what happens after the robot has stopped.

Appendix: computed coverage tables

Every figure below is produced by radar-compare/scripts/geometry.py from first principles, not copied from a vendor page. Re-run that script to reproduce them.

Beam footprint width

Full lateral width of the cone, $w = 2 d \tan(\text{FoV}/2)$.

Field of view	1 ft	3 ft	5 ft	8 ft	10 ft
15 deg	80 mm / 3.2 in	241 mm / 9.5 in	401 mm / 15.8 in	642 mm / 25.3 in	803 mm / 31.6 in
20 deg	108 mm / 4.2 in	322 mm / 12.7 in	537 mm / 21.2 in	860 mm / 33.9 in	1075 mm / 42.3 in
25 deg	135 mm / 5.3 in	405 mm / 16.0 in	676 mm / 26.6 in	1081 mm / 42.6 in	1352 mm / 53.2 in
27 deg	146 mm / 5.8 in	439 mm / 17.3 in	732 mm / 28.8 in	1171 mm / 46.1 in	1464 mm / 57.6 in
30 deg	163 mm / 6.4 in	490 mm / 19.3 in	817 mm / 32.2 in	1307 mm / 51.4 in	1633 mm / 64.3 in
45 deg	252 mm / 9.9 in	758 mm / 29.8 in	1262 mm / 49.7 in	2020 mm / 79.5 in	2525 mm / 99.4 in
55 deg	317 mm / 12.5 in	952 mm / 37.5 in	1587 mm / 62.5 in	2539 mm / 99.9 in	3173 mm / 124.9 in
60 deg	352 mm / 13.9 in	1056 mm / 41.6 in	1760 mm / 69.3 in	2816 mm / 110.9 in	3520 mm / 138.6 in
63 deg	374 mm / 14.7 in	1121 mm / 44.1 in	1868 mm / 73.5 in	2988 mm / 117.7 in	3736 mm / 147.1 in
65 deg	388 mm / 15.3 in	1165 mm / 45.9 in	1942 mm / 76.4 in	3107 mm / 122.3 in	3884 mm / 152.9 in
80 deg	512 mm / 20.1 in	1534 mm / 60.4 in	2558 mm / 100.7 in	4092 mm / 161.1 in	5115 mm / 201.4 in
90 deg	610 mm / 24.0 in	1829 mm / 72.0 in	3048 mm / 120.0 in	4877 mm / 192.0 in	6096 mm / 240.0 in
100 deg	726 mm / 28.6 in	2180 mm / 85.8 in	3632 mm / 143.0 in	5812 mm / 228.8 in	7265 mm / 286.0 in
110 deg	871 mm / 34.3 in	2612 mm / 102.8 in	4353 mm / 171.4 in	6965 mm / 274.2 in	8706 mm / 342.8 in
120 deg	1056 mm / 41.6 in	3168 mm / 124.7 in	5279 mm / 207.8 in	8447 mm / 332.6 in	10559 mm / 415.7 in
150 deg	2275 mm / 89.6 in	6825 mm / 268.7 in	11375 mm / 447.8 in	18200 mm / 716.6 in	22751 mm / 895.7 in
180 deg	unbounded	unbounded	unbounded	unbounded	unbounded

Sensors needed to close a 360 degree ring

Bare closure needs $n \times \text{FoV} \geq 360$. The overlap columns add the margin that tolerates mounting error and one dead sensor.

Field of view	Bare 360 deg	With 5 deg overlap	With 10 deg overlap	With 25 percent overlap
15 deg	24	36	72	32
20 deg	18	24	36	24
25 deg	15	18	24	20
27 deg	14	17	22	18
30 deg	12	15	18	16
45 deg	8	9	11	11
55 deg	7	8	8	9
60 deg	6	7	8	8
63 deg	6	7	7	8
65 deg	6	6	7	8

Field of view	Bare 360 deg	With 5 deg overlap	With 10 deg overlap	With 25 percent overlap
80 deg	5	5	6	6
90 deg	4	5	5	6
100 deg	4	4	4	5
110 deg	4	4	4	5
120 deg	3	4	4	4
150 deg	3	3	3	4
180 deg	2	3	3	3

Where the blind wedge between adjacent sensors closes

Measured outward from the platform edge, sensors on a 175 mm radius aimed radially. "never" means $n \times \text{FoV} < 360$, where no range ever closes the gap.

Sensors in the ring	15 deg	25 deg	27 deg	45 deg	63 deg	65 deg	90 deg	100 deg	120 deg
3	never	never	never	never	never	never	never	never	never
4	never	never	never	never	never	never	never	1363 mm	411 mm
5	never	never	never	never	never	never	616 mm	379 mm	198 mm
6	never	never	never	never	3318 mm	1981 mm	303 mm	217 mm	128 mm
8	never	never	never	never	410 mm	366 mm	148 mm	115 mm	74 mm
10	never	never	never	679 mm	217 mm	200 mm	98 mm	78 mm	52 mm
12	never	never	never	338 mm	147 mm	138 mm	72 mm	59 mm	39 mm
16	never	1561 mm	866 mm	168 mm	89 mm	84 mm	48 mm	39 mm	27 mm

Centre-zone footprint of the grid imagers

One zone of the array, with how many zones the subject spans across.

Grid imager	1 ft	3 ft	5 ft	8 ft	10 ft
VL53L5CX / VL53L8CX 4x4	187 mm (adult 2.41, cat 0.75)	560 mm (adult 0.8, cat 0.25)	934 mm (adult 0.48, cat 0.15)	1494 mm (adult 0.3, cat 0.09)	1868 mm (adult 0.24, cat 0.07)
VL53L5CX / VL53L8CX 8x8	93 mm (adult 4.82, cat 1.5)	280 mm (adult 1.61, cat 0.5)	467 mm (adult 0.96, cat 0.3)	747 mm (adult 0.6, cat 0.19)	934 mm (adult 0.48, cat 0.15)
VL53L7CX 4x4	305 mm (adult 1.48, cat 0.46)	914 mm (adult 0.49, cat 0.15)	1524 mm (adult 0.3, cat 0.09)	2438 mm (adult 0.18, cat 0.06)	3048 mm (adult 0.15, cat 0.05)
VL53L7CX 8x8	152 mm (adult 2.95, cat 0.92)	457 mm (adult 0.98, cat 0.31)	762 mm (adult 0.59, cat 0.18)	1219 mm (adult 0.37, cat 0.11)	1524 mm (adult 0.3, cat 0.09)
AMG8833 Grid-EYE 8x8	88 mm (adult 5.11, cat 1.59)	264 mm (adult 1.7, cat 0.53)	440 mm (adult 1.02, cat 0.32)	704 mm (adult 0.64, cat 0.2)	880 mm (adult 0.51, cat 0.16)
MLX90640 32x24 (55 deg)	20 mm (adult 22.69, cat 7.06)	60 mm (adult 7.56, cat 2.35)	99 mm (adult 4.54, cat 1.41)	159 mm (adult 2.84, cat 0.88)	198 mm (adult 2.27, cat 0.71)
MLX90640 32x24 (110 deg)	54 mm (adult 8.27, cat 2.57)	163 mm (adult 2.76, cat 0.86)	272 mm (adult 1.65, cat 0.51)	435 mm (adult 1.03, cat 0.32)	544 mm (adult 0.83, cat 0.26)
MLX90641 16x12 (55 deg)	40 mm (adult 11.34, cat 3.53)	119 mm (adult 3.78, cat 1.18)	198 mm (adult 2.27, cat 0.71)	317 mm (adult 1.42, cat 0.44)	397 mm (adult 1.13, cat 0.35)
TMF8821 3x3	120 mm (adult 3.74, cat 1.16)	361 mm (adult 1.25, cat 0.39)	602 mm (adult 0.75, cat 0.23)	963 mm (adult 0.47, cat 0.15)	1204 mm (adult 0.37, cat 0.12)
TMF8821 4x4	123 mm (adult 3.65, cat 1.14)	369 mm (adult 1.22, cat 0.38)	616 mm (adult 0.73, cat 0.23)	985 mm (adult 0.46, cat 0.14)	1232 mm (adult 0.37, cat 0.11)

Vertical coverage, no down-tilt

Y means part of the subject is inside the beam at 1 / 3 / 5 / 8 / 10 ft; a dot means it is missed entirely. Floor strike is where the beam first reaches the floor.

Mount height	Vertical FoV	Floor strike	adult human standing	toddler	large dog	small dog	cat	cat lying	power cable	table top
100 mm	25 deg	451 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	Y
100 mm	45 deg	241 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	..YYY
100 mm	63 deg	163 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	..YYY
100 mm	90 deg	100 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY
200 mm	25 deg	902 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	...YY
200 mm	45 deg	483 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	..YYY
200 mm	63 deg	326 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY
200 mm	90 deg	200 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY
400 mm	25 deg	1804 mm	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	..YYY	...YY	..YYY
400 mm	45 deg	966 mm	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	.YYYY	..YYY	.YYYY
400 mm	63 deg	653 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	.YYYY
400 mm	90 deg	400 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	YYYYY
800 mm	25 deg	3609 mm	YYYYY	YYYYY	.YYYY	...YY	Y	Y		YYYYY
800 mm	45 deg	1931 mm	YYYYY	YYYYY	.YYYY	..YYY	..YYY	..YY	...YY	YYYYY
800 mm	63 deg	1306 mm	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	..YYY	..YYY	YYYYY
800 mm	90 deg	800 mm	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	.YYYY	.YYYY	YYYYY

Vertical coverage, 15 degree down-tilt

Same, sensor tilted down.

Mount height	Vertical FoV	Floor strike	adult human standing	toddler	large dog	small dog	cat	cat lying	power cable	table top
100 mm	25 deg	192 mm	YYY..	YYY..	YYY..	YYY..	YYY..	YYY..	YYY..	
100 mm	45 deg	130 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	
100 mm	63 deg	95 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	...YY
100 mm	90 deg	58 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	..YYY
200 mm	25 deg	384 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	
200 mm	45 deg	261 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	
200 mm	63 deg	190 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	...YY
200 mm	90 deg	116 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY
400 mm	25 deg	768 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	
400 mm	45 deg	521 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	...YY
400 mm	63 deg	380 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	..YYY
400 mm	90 deg	231 mm	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY
800 mm	25 deg	1537 mm	YYYYY	YYYYY	YYYYY	.YYYY	..YYY	..YYY	...YY	YYY..
800 mm	45 deg	1043 mm	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	.YYYY	..YYY	YYYYY
800 mm	63 deg	759 mm	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	.YYYY	.YYYY	YYYYY
800 mm	90 deg	462 mm	YYYYY	YYYYY	YYYYY	YYYYY	.YYYY	.YYYY	.YYYY	YYYYY

Stopping distance against sensor update rate

Reaction distance is speed x (sensor period + 30 ms of compute). An obstacle closer than the total must never be the first time the robot sees it.

Speed	Sensor and update rate	Sensor lag	Braking	Reaction distance	Braking distance	Total stop
0.3 m/s	VL53L1X single zone @ 50 Hz	20.0 ms	firm 0.5 g	15 mm	9 mm	24 mm / 1.0 in
0.3 m/s	VL53L5CX 4x4 @ 60 Hz	16.7 ms	firm 0.5 g	14 mm	9 mm	23 mm / 0.9 in
0.3 m/s	VL53L5CX 8x8 @ 15 Hz	66.7 ms	firm 0.5 g	29 mm	9 mm	38 mm / 1.5 in
0.3 m/s	VL53L8CX 8x8 @ 15 Hz	66.7 ms	firm 0.5 g	29 mm	9 mm	38 mm / 1.5 in
0.3 m/s	typical 24 GHz mmWave @ 10 Hz	100.0 ms	firm 0.5 g	39 mm	9 mm	48 mm / 1.9 in
0.3 m/s	AMG8833 thermal @ 10 Hz	100.0 ms	firm 0.5 g	39 mm	9 mm	48 mm / 1.9 in
0.3 m/s	AMG8833 thermal @ 1 Hz	1000.0 ms	firm 0.5 g	309 mm	9 mm	318 mm / 12.5 in
0.3 m/s	MLX90640 32x24 @ 8 Hz	125.0 ms	firm 0.5 g	46 mm	9 mm	56 mm / 2.2 in
0.3 m/s	RPLIDAR A1 full scan @ 5.5 Hz	181.8 ms	firm 0.5 g	64 mm	9 mm	73 mm / 2.9 in
0.5 m/s	VL53L1X single zone @ 50 Hz	20.0 ms	firm 0.5 g	25 mm	26 mm	50 mm / 2.0 in
0.5 m/s	VL53L5CX 4x4 @ 60 Hz	16.7 ms	firm 0.5 g	23 mm	26 mm	49 mm / 1.9 in
0.5 m/s	VL53L5CX 8x8 @ 15 Hz	66.7 ms	firm 0.5 g	48 mm	26 mm	74 mm / 2.9 in
0.5 m/s	VL53L8CX 8x8 @ 15 Hz	66.7 ms	firm 0.5 g	48 mm	26 mm	74 mm / 2.9 in
0.5 m/s	typical 24 GHz mmWave @ 10 Hz	100.0 ms	firm 0.5 g	65 mm	26 mm	90 mm / 3.6 in
0.5 m/s	AMG8833 thermal @ 10 Hz	100.0 ms	firm 0.5 g	65 mm	26 mm	90 mm / 3.6 in
0.5 m/s	AMG8833 thermal @ 1 Hz	1000.0 ms	firm 0.5 g	515 mm	26 mm	540 mm / 21.3 in
0.5 m/s	MLX90640 32x24 @ 8 Hz	125.0 ms	firm 0.5 g	78 mm	26 mm	103 mm / 4.1 in
0.5 m/s	RPLIDAR A1 full scan @ 5.5 Hz	181.8 ms	firm 0.5 g	106 mm	26 mm	131 mm / 5.2 in
1.0 m/s	VL53L1X single zone @ 50 Hz	20.0 ms	firm 0.5 g	50 mm	102 mm	152 mm / 6.0 in
1.0 m/s	VL53L5CX 4x4 @ 60 Hz	16.7 ms	firm 0.5 g	47 mm	102 mm	149 mm / 5.9 in
1.0 m/s	VL53L5CX 8x8 @ 15 Hz	66.7 ms	firm 0.5 g	97 mm	102 mm	199 mm / 7.8 in
1.0 m/s	VL53L8CX 8x8 @ 15 Hz	66.7 ms	firm 0.5 g	97 mm	102 mm	199 mm / 7.8 in
1.0 m/s	typical 24 GHz mmWave @ 10 Hz	100.0 ms	firm 0.5 g	130 mm	102 mm	232 mm / 9.1 in
1.0 m/s	AMG8833 thermal @ 10 Hz	100.0 ms	firm 0.5 g	130 mm	102 mm	232 mm / 9.1 in
1.0 m/s	AMG8833 thermal @ 1 Hz	1000.0 ms	firm 0.5 g	1030 mm	102 mm	1132 mm / 44.6 in
1.0 m/s	MLX90640 32x24 @ 8 Hz	125.0 ms	firm 0.5 g	155 mm	102 mm	257 mm / 10.1 in
1.0 m/s	RPLIDAR A1 full scan @ 5.5 Hz	181.8 ms	firm 0.5 g	212 mm	102 mm	314 mm / 12.4 in